

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	Slot A	Date:	14 - 08 - 2026

Code of Conduct

*I **Mohammed Rayan Shaikh - 192424059** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

- Analyze the semantic representations and identify the action-object relationships.
 - Determine which query contains semantic interpretation errors.
 - Evaluate how meaning representation affects chatbot decision accuracy.
 - Recommend improvements to enhance semantic understanding in telecom customer support systems.
2. First-Order Predicate Calculus for Smart Manufacturing.
An Industry 4.0 manufacturing plant monitors machines using semantic rules.
Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
 2. Analyze which products are currently available.
 3. Infer whether Gear production is affected by maintenance activities.
 4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.
3. Word Sense Disambiguation in E-Commerce Search Engines.
An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
 2. Identify the semantic cues used for disambiguation.
 3. Evaluate the impact of incorrect sense selection on search engine performance.
 4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.
4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.
A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Semantic Representation in Customer Support Chatbots

Given Semantic Representations

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

1. Analyze the semantic representations and identify the action-object relationships.

The semantic representation follows the general form:

ACTION(Object, Customer)

where the first element represents the action and the second element represents the customer performing or requesting the action.

- **ACTIVATE(Roaming, Customer)**
Action = ACTIVATE
Object = Roaming
Relationship = Customer wants to activate international roaming.
- **DEACTIVATE(Caller_Tune, Customer)**
Action = DEACTIVATE
Object = Caller Tune
Relationship = Customer wants to deactivate the caller tune service.
- **QUERY(Data_Balance, Customer)**
Action = QUERY
Object = Data Balance
Relationship = Customer wants to check the available data balance.
- **ACTIVATE(5G_Service, Customer)**
Action = ACTIVATE
Object = 5G Service
Relationship = Customer wants to enable 5G service.

Thus, the action-object relationship correctly represents the main intention expressed in each query.

2. Determine which query contains semantic interpretation errors.

From the chatbot logs:

Query ID	Actual Intent	Predicted Intent	Result
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Deactivate Caller Tune	Correct
Q3	Check Data Balance	Query Data Balance	Correct
Q4	Enable 5G Service	Activate 5G Service	Correct

Strictly according to the provided semantic representations, **none of the four queries contains a major semantic interpretation error.**

Q3 uses the word "QUERY" instead of "CHECK", but both represent the same user intention. Similarly, "Enable" and "Activate" have equivalent meanings in the context of 5G service.

Therefore, the semantic analyzer has correctly identified the intended actions and objects.

3. Evaluate how meaning representation affects chatbot decision accuracy.

Meaning representation directly affects chatbot decision-making because it converts natural-language queries into structured concepts.

For example:

"Activate international roaming for my number."

is converted into:

ACTIVATE(Roaming, Customer)

This representation makes it easier for the chatbot to identify the required service operation.

If the system incorrectly represents:

"Deactivate caller tune service."

as:

ACTIVATE(Caller_Tune, Customer)

the chatbot could perform the opposite operation, resulting in an incorrect response or service action.

Similarly, confusing:

QUERY(Data_Balance, Customer)

with another service operation could result in an incorrect response.

Therefore, accurate semantic representations are essential for reliable intent detection, service selection, and response generation.

4. Recommendations to improve semantic understanding in telecom customer support systems.

The following improvements can be implemented:

1. Use domain-specific telecom vocabulary such as roaming, caller tune, data balance, 5G, recharge, and SIM.

2. Use intent classification models trained on real customer queries.
3. Maintain separate representations for actions and objects.
4. Include synonyms such as "activate", "enable", and "turn on".
5. Handle negation carefully, such as "do not activate roaming".
6. Use contextual information from previous messages.
7. Use entity recognition to identify services, phone numbers, plans, and account information.
8. Continuously evaluate the system using actual chatbot logs.
9. Use confidence scores to detect uncertain predictions.
10. Use human review for low-confidence service requests.

Conclusion

Semantic representation provides a structured connection between a customer's natural-language query and the action that the chatbot must perform. Accurate action-object relationships improve intent recognition and reduce incorrect service operations.

2. First-Order Predicate Calculus for Smart Manufacturing

Production Data

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules

1. $\text{Active}(x) \rightarrow \text{Producing}(x)$
2. $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
3. $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

1. Represent the production data using First-Order Predicate Calculus.

The production data can be represented using the following facts:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

Applying the first rule:

Active(x) $\rightarrow$ Producing(x)

we obtain:

Producing(M1)

Producing(M2)

Producing(M4)

From the third rule:

Maintenance(x) $\rightarrow$ $\neg$ Producing(x)

we obtain:

$\neg$ Producing(M3)

Therefore, the complete representation is:

- Active(M1)
- Active(M2)
- Active(M4)
- Maintenance(M3)

- Producing(M1)
- Producing(M2)
- Producing(M4)
- $\neg$ Producing(M3)

2. Analyze which products are currently available.

The second rule is:

Produces(x,y) $\wedge$ Active(x) $\rightarrow$ Available(y)

To determine that a product is available, we need two facts:

1. A machine produces the product.
2. The machine is active.

However, the given production data does not specify any **Produces(x,y)** facts.

Therefore, no specific product can be logically identified as available from the given information.

For example, if we were given:

Produces(M1,Gear)

and:

Active(M1)

then we could infer:

Available(Gear)

because:

Produces(M1,Gear) $\wedge$ Active(M1) $\rightarrow$ Available(Gear)

Hence, based strictly on the supplied data, **no product availability can be determined.**

3. Infer whether Gear production is affected by maintenance activities.

The data shows:

Maintenance(M3)

Using the rule:

Maintenance(x) $\rightarrow$ $\neg$ Producing(x)

we infer:

$\neg$ Producing(M3)

Therefore, machine M3 cannot be considered a producing machine while it is under maintenance.

If M3 produces Gear, then Gear production through M3 would be affected.

For example:

Produces(M3,Gear)

Maintenance(M3)

would imply:

$\neg$ Producing(M3)

Therefore, M3 cannot currently produce Gear.

However, because the given dataset does not explicitly state that M3 produces Gear, we cannot conclude that total Gear production is affected.

4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

First-Order Predicate Calculus is useful because it represents machines, products, statuses, and relationships formally.

For example:

Active(M1) $\rightarrow$ Producing(M1)

allows the system to automatically infer production status.

Advantages include:

- Clear representation of manufacturing facts.
- Automatic inference from predefined rules.
- Easy identification of machine conditions.
- Support for maintenance decision-making.
- Ability to represent relationships between machines and products.
- Reduced ambiguity in decision rules.

However, predicate logic depends heavily on the completeness and accuracy of the available facts and rules. It cannot infer information that has not been provided.

Conclusion

First-Order Predicate Calculus provides a formal framework for representing machine states and deriving production-related conclusions. It is particularly useful for rule-based industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines

Given Search Logs

User Query	Possible Sense 1	Possible Sense 2
Apple accessories	Fruit	Technology Brand
Mouse wireless	Animal	Computer Device
Java tutorial	Island	Programming Language
Python course	Snake	Programming Language

Search Context

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

1. Analyze the contextual information and determine the correct word sense.

a. "Apple accessories"

Possible meanings:

- Apple = fruit
- Apple = technology brand

The clicked result is **iPhone Charger**.

An iPhone is a product of the technology company Apple.

Therefore:

Correct sense = Technology Brand

b. "Mouse wireless"

Possible meanings:

- Mouse = animal
- Mouse = computer device

The clicked result is **Bluetooth Mouse**.

Therefore:

Correct sense = Computer Device

c. "Java tutorial"

Possible meanings:

- Java = island
- Java = programming language

The word "tutorial" and the clicked result "Coding Lessons" strongly indicate programming.

Therefore:

Correct sense = Programming Language

d. "Python course"

Possible meanings:

- Python = snake
- Python = programming language

The clicked result is **Software Development Training**.

Therefore:

Correct sense = Programming Language

Correct Sense Table

Query	Correct Sense	Contextual Evidence
Apple accessories	Technology Brand	iPhone Charger
Mouse wireless	Computer Device	Bluetooth Mouse
Java tutorial	Programming Language	Coding Lessons
Python course	Programming Language	Software Development Training

2. Identify semantic cues used for disambiguation.

The important contextual cues are:

- **Apple + accessories + iPhone Charger** → Technology Brand
- **Mouse + wireless + Bluetooth Mouse** → Computer Device
- **Java + tutorial + Coding Lessons** → Programming Language
- **Python + course + Software Development Training** → Programming Language

The surrounding words provide information about the intended meaning of an ambiguous word.

3. Impact of incorrect sense selection on search engine performance.

Incorrect word sense selection can significantly reduce search relevance.

For example, if "Apple accessories" is interpreted as fruit, the system could return:

- Fruit baskets
- Apple recipes
- Fruit storage products

instead of:

- iPhone chargers
- iPhone cases
- Apple Watch accessories

Similarly, interpreting "Java tutorial" as an island could produce travel-related results instead of programming tutorials.

This can lead to:

- Lower search relevance.
- Lower click-through rate.
- Poor recommendations.
- Reduced customer satisfaction.
- Incorrect product ranking.
- Lower conversion rates.

4. Industrial-scale WSD strategy for improving recommendation accuracy.

An industrial-scale e-commerce system should combine multiple sources of information:

1. **Contextual word embeddings** to understand surrounding words.
2. **User search history** to identify user interests.
3. **Clicked-product information** to infer intended meaning.
4. **Product categories** to distinguish different senses.
5. **Query expansion** using related terms.
6. **Knowledge graphs** linking products and concepts.
7. **Machine-learning-based WSD** trained using search and click data.
8. **User personalization** based on previous interactions.
9. **Feedback mechanisms** using clicks, purchases, and dwell time.
10. **Continuous evaluation** using search relevance metrics.

Conclusion

Word Sense Disambiguation is essential in e-commerce because many words have multiple meanings. Context from the query and user interaction can help determine the correct meaning and improve search ranking and product recommendations.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

Parsed Sentences

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Generated Semantic Roles

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

1. Analyze how syntactic structures contribute to semantic role assignment.

Syntax determines the relationship between entities and actions in a sentence.

For example:

"Doctor prescribed medicine to patient."

The structure is:

Subject → Verb → Object → Recipient

Therefore:

- Doctor = Agent
- Prescribed = Action
- Medicine = Object/Theme
- Patient = Recipient

Similarly:

"Nurse monitored patient continuously."

The syntactic structure identifies:

- Nurse = Agent
- Monitored = Action
- Patient = Object/Theme

Thus, syntactic parsing helps the system determine who performed an action, what was affected, and who received something.

2. Determine whether the assigned semantic roles are appropriate.

The given semantic roles are only partially appropriate.

Doctor → Agent

This is correct.

In:

"Doctor prescribed medicine to patient."

the doctor performs the action.

Medicine → Instrument

This is not the most appropriate role.

Medicine is the **Theme/Object** being prescribed, rather than an instrument.

A better representation is:

Medicine → Theme

Patient → Recipient

This is correct in:

"Doctor prescribed medicine to patient."

The patient receives the medicine.

However, in:

"Patient reported severe headache."

the patient is better represented as an **Experiencer/Agent**, while headache is the **Symptom**.

Headache → Symptom

This is correct.

Corrected Semantic Roles

Entity	Correct Role
Doctor	Agent
Medicine	Theme/Object
Patient	Recipient / Experiencer / Patient
Headache	Symptom

Sentence-Level Representation

Doctor prescribed medicine to patient.

Can be represented as:

PRESCRIBE(Doctor, Medicine, Patient)

where:

- Doctor = Agent
- Medicine = Theme
- Patient = Recipient

Patient reported severe headache.

Can be represented as:

REPORT(Patient, Headache)

where:

- Patient = Experiencer/Agent
- Headache = Symptom

Nurse monitored patient continuously.

Can be represented as:

MONITOR(Nurse, Patient)

where:

- Nurse = Agent
- Patient = Theme/Patient

Medicine reduced blood pressure.

Can be represented as:

REDUCE(Medicine, Blood_Pressure)

where:

- Medicine = Cause/Agent
- Blood Pressure = Theme/Condition

3. Evaluate potential errors from incorrect parsing.

Incorrect syntactic or semantic parsing can produce serious errors in healthcare systems.

For example, confusing the roles in:

"Doctor prescribed medicine to patient."

could lead the system to interpret the patient as the person prescribing the medicine.

Similarly, treating **medicine as an instrument** instead of a theme can result in an incorrect representation of the medical event.

Potential consequences include:

- Incorrect medical information extraction.
- Incorrect patient records.
- Wrong medication-event relationships.
- Incorrect clinical decision support.
- Errors in medical question answering.
- Reduced accuracy in medical information retrieval.

Therefore, semantic role assignment must be carefully validated.

4. Methods to improve syntax-driven semantic analysis.

The healthcare NLP system can be improved using:

1. **Dependency parsing** to identify grammatical relationships.
2. **Semantic Role Labeling (SRL)** to assign Agent, Theme, Recipient, Experiencer, and other roles.
3. **Medical Named Entity Recognition** to identify diseases, medicines, symptoms, and patients.
4. **Medical ontologies** such as SNOMED CT and UMLS for standardized concepts.
5. **Domain-specific language models** trained on medical text.

Conclusion

Syntax-driven semantic analysis helps healthcare systems convert grammatical structures into meaningful medical relationships. Combining syntactic parsing, semantic role labeling, medical entity recognition, and domain knowledge can significantly improve the accuracy of healthcare information systems.